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EXP = 2

TITLE = QUICK SORT

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In [18]: import random
import time
import matplotlib.pyplot as plt

def quick_sort(arr):
    if len(arr) <= 1:
        return arr
    pivot = arr[len(arr) // 2]
    left = [x for x in arr if x < pivot]
    middle = [x for x in arr if x == pivot]
    right = [x for x in arr if x > pivot]
    return quick_sort(left) + middle + quick_sort(right)

arr = list(map(int, input("Enter elements separated by space: ").split()))
print("Input array: ", arr)

sorted_arr = quick_sort(arr)
print("Sorted array:", sorted_arr)

start_time = time.time()
sorted_arr = quick_sort(arr)
end_time = time.time()
print("Time taken to sort:", end_time - start_time, "seconds")
print("=====")

n_values = [5000, 6000, 7000, 8000]
time_values = []

for n in n_values:
    arr = [random.randint(1, 9) for _ in range(n)]

    start_time = time.time()
    sorted_arr = quick_sort(arr)
    end_time = time.time()

    time_taken = end_time - start_time
    print("Time taken to sort", n, "elements:", time_taken, "seconds")
    time_values.append(time_taken)
print("=====")

plt.plot(n_values, time_values, color = 'red', marker='o')
plt.xlabel('Number of elements (n)')
plt.ylabel('Time taken (seconds)')
plt.title('quick sort time complexity analysis')
plt.grid(True)
plt.show()
```

Input array: [5000, 6000, 7000, 8000, 9000, 1000]

Sorted array: [1000, 5000, 6000, 7000, 8000, 9000]

Time taken to sort: 0.0 seconds

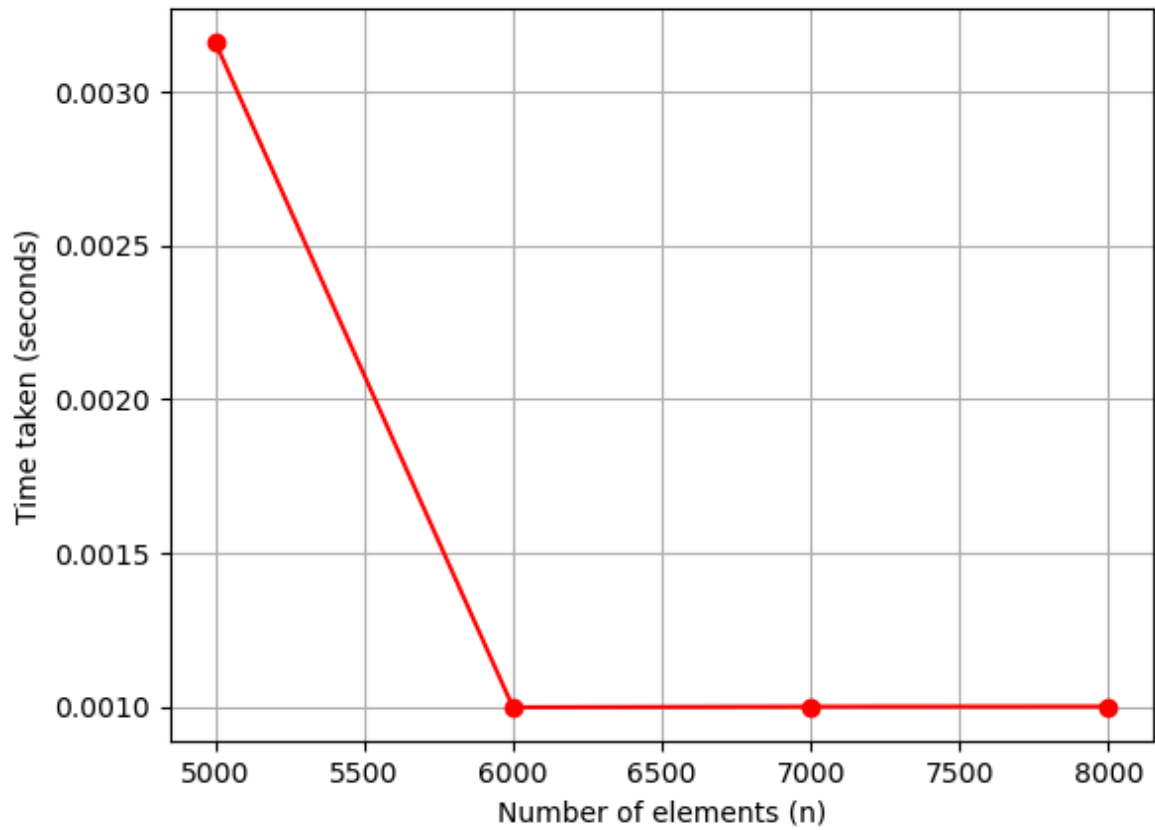
Time taken to sort 5000 elements: 0.003162384033203125 seconds

Time taken to sort 6000 elements: 0.000997304916381836 seconds

Time taken to sort 7000 elements: 0.00099945068359375 seconds

Time taken to sort 8000 elements: 0.0009999275207519531 seconds

quick sort time complexity analysis



In []: